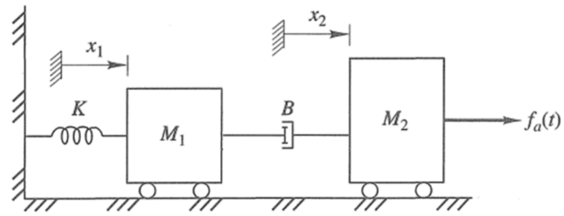
MENG 3020 – Quiz 1 Solution – Fall 2025

Question A translational mechanical system is shown below. Spring is undeflected when $x_1 = x_2 = 0$. The *system input* is the applied force $f_a(t)$ and the *system output* is the displacement of spring K.



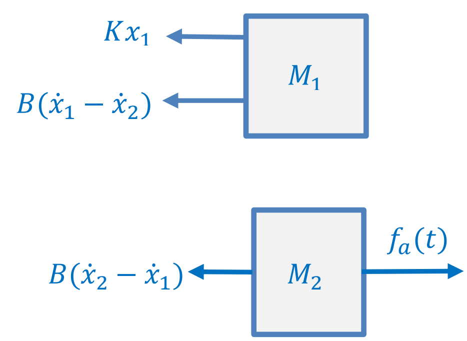
Solve the following questions:

a) Draw the free-body diagrams for mass M_1 and M_2. Show all forces applied and write a set of ordinary differential equations of motion. Show your work.

From the free-body diagrams and applying Newton's second law:

Mass M_1 $\rightarrow$ $\quad -Kx_1 - B(\dot{x}_1 - \dot{x}_2) = M_1\ddot{x}_1$

Mass M_2 $\rightarrow$ $\quad f_a(t) - B(\dot{x}_2 - \dot{x}_1) = M_2\ddot{x}_2$



b) Complete the following block diagram model based on the equations of motion in Part (a).

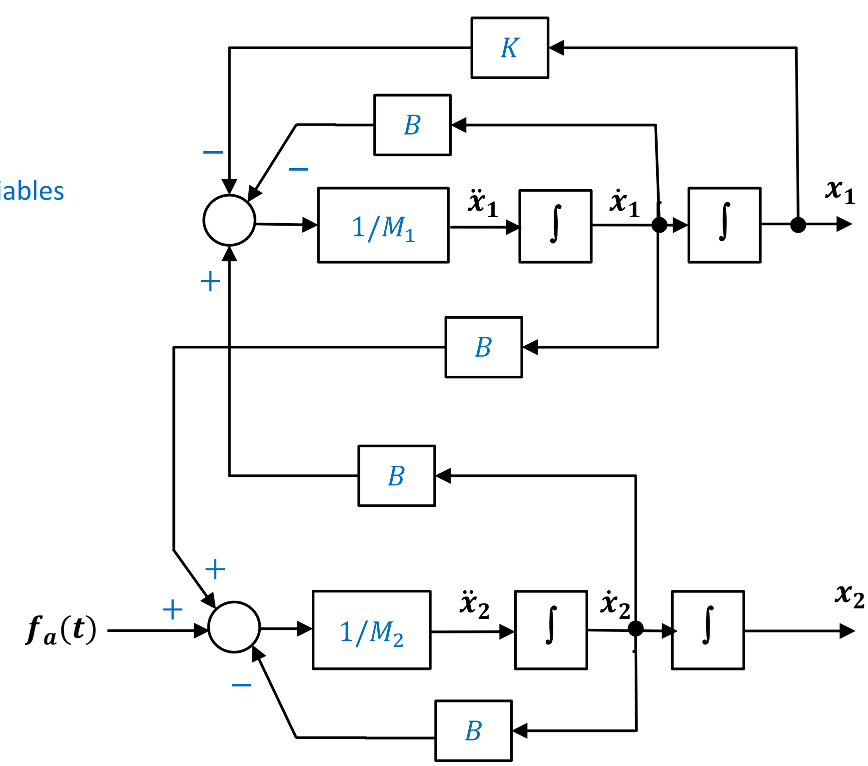
The equations of motion are:

$$\begin{cases} M_1\ddot{x}_1 + B\dot{x}_1 + Kx_1 - B\dot{x}_2 = 0 \\ M_2\ddot{x}_2 + B\dot{x}_2 - B\dot{x}_1 = f_a(t) \end{cases}$$

Solve for highest derivative of output variables

$$\ddot{x}_1 = \frac{1}{M_1}(-B\dot{x}_1 - Kx_1 + B\dot{x}_2)$$

$$\dot{x}_2 = \frac{1}{M_2}(f_a(t) - B\dot{x}_2 + B\dot{x}_1)$$



c) Develop the *state-variable equations* and the *output equation* for the obtained equations of motion in Part (a). Write the state-space equations in *matrix-vector* form.

$$\begin{cases} \dot{\mathbf{q}}(\boldsymbol{t}) = \boldsymbol{A}\mathbf{q}(\boldsymbol{t}) + \boldsymbol{B}\mathbf{u}(\boldsymbol{t}) \\ \mathbf{y}(\boldsymbol{t}) = \boldsymbol{C}\mathbf{q}(\boldsymbol{t}) + \boldsymbol{D}\mathbf{u}(\boldsymbol{t}) \end{cases}$$

The state variables are selected as the displacement of the spring K and velocity of the mass M_1 and M_2 . Show your work.

$$q_1 = x_1, \qquad q_2 = \dot{x}_1, \qquad q_3 = \dot{x}_2$$

$$q_1 = x_1 \ \rightarrow \ \dot{q}_1 = \dot{x}_1 \ \rightarrow \ \dot{q}_1 = q_2 \qquad \text{Eqn. } (1)$$

$$q_2 = \dot{x}_1 \ \rightarrow \ \dot{q}_2 = \ddot{x}_1 \ \rightarrow \ \dot{q}_2 = \frac{1}{M_1}(-B\dot{x}_1 - Kx_1 + B\dot{x}_2) \ \rightarrow \ \dot{q}_2 = \frac{1}{M_1}(-Bq_2 - Kq_1 + Bq_3) \qquad \text{Eqn. } (2)$$

$$q_3 = \dot{x}_2 \ \rightarrow \ \dot{q}_3 = \ddot{x}_2 \ \rightarrow \ \dot{q}_3 = \frac{1}{M_2}(f_a(t) - B\dot{x}_2 + B\dot{x}_1) \ \rightarrow \ \dot{q}_3 = \frac{1}{M_2}(f_a(t) - Bq_3 + Bq_2) \qquad \text{Eqn. } (3)$$

The state-variable equations and the output equation are obtained as:

$$\dot{q}_1 = q_2$$
$$\dot{q}_2 = \frac{1}{M_1}(-Bq_2 - Kq_1 + Bq_3)$$
$$\dot{q}_3 = \frac{1}{M_2}(f_a(t) - Bq_3 + Bq_2)$$

$$y = x_1 \ \rightarrow \ y = q_1$$

The state equation and output equation in the standard matrix-vector form are:

State Equation: $\dot{\mathbf{q}}(\boldsymbol{t}) = \boldsymbol{A}\mathbf{q}(\boldsymbol{t}) + \boldsymbol{B}\mathbf{u}(\boldsymbol{t})$

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -K/M_1 & -B/M_1 & B/M_1 \\ 0 & B/M_2 & -B/M_2 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1/M_2 \end{bmatrix} f_a(t)$$

Output Equation: $\mathbf{y}(\boldsymbol{t}) = \boldsymbol{C}\mathbf{q}(\boldsymbol{t}) + \boldsymbol{D}\mathbf{u}(\boldsymbol{t})$

$$y(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + [0]u(t)$$